心跳脈搏與年齡健康指南

A Guide to Heart Rate, Pulse, and Age-Related Health

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引言 (Introduction)

心率，或稱脈搏，是衡量心血管健康狀況的關鍵生命體徵之一。它指的是心臟每分鐘跳動的次數（beats per minute, bpm）。無論是靜息狀態還是運動期間，監測心率都有助於評估個人的健康水平、體能狀況，並及早發現潛在的健康問題。本報告旨在全面探討心率的各個方面，包括不同年齡層的正常靜息心率範圍、運動時的目標心率區間，以及心率異常的潛在風險與改善策略，為您提供一份詳盡的雙語健康參考。

Heart rate, or pulse, is one of the key vital signs for measuring cardiovascular health. It refers to the number of times the heart beats per minute (bpm). Monitoring heart rate, both at rest and during exercise, helps assess an individual's overall health, fitness level, and can aid in the early detection of potential health issues. This report aims to provide a comprehensive overview of various aspects of heart rate, including normal resting heart rate ranges for different age groups, target heart rate zones for exercise, and the potential risks and improvement strategies for abnormal heart rates, offering a detailed bilingual health reference.

靜息心率 (Resting Heart Rate, RHR)

靜息心率是指在清醒、不活動的安靜狀態下，心臟每分鐘跳動的次數。它是評估心臟功能和整體健康狀況的一個重要指標。一般而言，較低的靜息心率意味著心肌功能更有效率，心血管健康狀況更佳。正常成年人的靜息心率範圍為每分鐘60至100次 [1]。然而，這個範圍會因年齡、體能水平及其他因素而異。

Resting Heart Rate (RHR) is the number of times your heart beats per minute when you are awake and in a state of rest. It is a crucial indicator for assessing cardiac function and overall health. Generally, a lower RHR implies more efficient heart muscle function and better cardiovascular fitness. The normal RHR for an adult is between 60 and 100 bpm [1]. However, this range can vary depending on age, fitness level, and other factors.

各年齡層的正常靜息心率 (Normal Resting Heart Rate by Age)

不同年齡階段的靜息心率標準有所不同，尤其是從嬰兒期到成年期的變化較為顯著。以下是根據克里夫蘭醫學中心提供的數據整理的各年齡層靜息心率參考範圍 [1]：

The standard for RHR differs across various life stages, with significant changes observed from infancy to adulthood. The following table, based on data from the Cleveland Clinic, outlines the reference ranges for RHR by age [1]:

年齡 (Age)	正常靜息心率 (Normal Resting Heart Rate)
新生兒 (0-4週) (Newborn, 0-4 weeks)	100 – 205 bpm
嬰兒 (4週-1歲) (Infant, 4 weeks-1 year)	100 – 180 bpm
幼兒 (1-3歲) (Toddler, 1-3 years)	98 – 140 bpm
學齡前兒童 (3-5歲) (Preschool, 3-5 years)	80 – 120 bpm
學齡兒童 (5-12歲) (School-age, 5-12 years)	75 – 118 bpm
青少年 (13-17歲) (Adolescent, 13-17 years)	60 – 100 bpm
成人 (18歲以上) (Adult, 18+ years)	60 – 100 bpm

靜息心率與體能水平 (Resting Heart Rate and Fitness Level)

靜息心率也反映了個人的體能狀況。經過規律訓練的運動員或體能較佳者，其心臟每次搏動能輸出更多血液，因此在靜息狀態下心跳次數會較少，通常介於40至60 bpm之間。以下是根據美國疾病管制與預防中心(CDC)數據整理的成年男女靜息心率與體能水平對照表 [2]：

RHR also reflects an individual's physical fitness level. For well-trained athletes or individuals with high fitness levels, the heart can pump more blood with each beat, resulting in a lower heart rate at rest, often between 40 and 60 bpm. The following tables, based on data from the U.S. Centers for Disease Control and Prevention (CDC), show the correlation between RHR and fitness levels for adult men and women [2]:

男性靜息心率對照表 (bpm)

Men's Resting Heart Rate Chart (bpm)

年齡 (Age)	運動員 (Athlete)	優秀 (Excellent)	良好 (Good)	平均 (Average)	較差 (Poor)
18-25	49-55	56-61	62-65	70-73	> 81

26-35	49-54	55-61	62-65	71-74	> 81
36-45	50-56	57-62	63-66	71-75	> 82
46-55	50-57	58-63	64-67	72-76	> 83
56-65	51-56	57-61	62-67	72-75	> 81
65+	50-55	56-61	62-65	70-73	> 79

女性靜息心率對照表 (bpm)

Women's Resting Heart Rate Chart (bpm)

年齡 (Age)	運動員 (Athlete)	優秀 (Excellent)	良好 (Good)	平均 (Average)	較差 (Poor)
18-25	54-60	61-65	66-69	74-78	> 84
26-35	54-59	60-64	65-68	73-76	> 82
36-45	54-59	60-64	65-69	74-78	> 84
46-55	54-60	61-65	66-69	74-77	> 83
56-65	54-59	60-64	65-68	74-77	> 83
65+	54-59	60-64	65-68	73-76	> 83

運動目標心率 (Target Heart Rate During Exercise)

在進行體育鍛煉時，維持在特定的心率區間內，可以最有效地達到健身目標，無論是減脂、提升耐力還是增強心肺功能。這個理想的心率範圍被稱為「目標心率區間」。

Maintaining your heart rate within a specific range during physical activity is the most effective way to achieve fitness goals, whether it's fat loss, endurance improvement, or cardiovascular enhancement. This ideal heart rate range is known as the "target heart rate zone."

如何計算最大心率與目標心率區間 (How to Calculate Max Heart Rate and Target Zone)

計算目標心率的第一步是估算您的「最大心率」（Maximum Heart Rate, MHR）。一個廣泛使用的簡易公式是 [3]：

The first step in calculating your target heart rate is to estimate your Maximum Heart Rate (MHR). A widely used simple formula is [3]:

> **最大心率 (MHR) = 220 - 年齡 (Your Age)**

例如，一位40歲的成年人，其預估最大心率為 `220 - 40 = 180 bpm`。

For example, a 40-year-old adult has a predicted MHR of `220 - 40 = 180 bpm`.

美國心臟協會 (AHA) 建議，中等強度運動的目標心率應為最大心率的 **50% 至 70%**，而高強度運動則應在最大心率的 **70% 至 85%** 之間 [3]。

The American Heart Association (AHA) recommends a target heart rate of **50% to 70%** of MHR for moderate-intensity exercise and **70% to 85%** for vigorous-intensity exercise [3].

以下是根據AHA數據整理的各年齡層運動目標心率區間參考表 [3]：

The following table, based on AHA data, provides a reference for target heart rate zones by age [3]:

年齡 (Age)	目標心率區間 (50-85%) (Target HR Zone)	預估最大心率 (100%) (Predicted Max HR)
20	100–170 bpm	200 bpm
30	95–162 bpm	190 bpm
35	93–157 bpm	185 bpm
40	90–153 bpm	180 bpm
45	88–149 bpm	175 bpm
50	85–145 bpm	170 bpm
55	83–140 bpm	165 bpm
60	80–136 bpm	160 bpm
65	78–132 bpm	155 bpm
70	75–128 bpm	150 bpm

運動強度心率區間 (Heart Rate Zones for Exercise Intensity)

根據最大心率的百分比，可以將運動強度劃分為五個區間，以對應不同的訓練目的 [4]：

Based on the percentage of MHR, exercise intensity can be divided into five zones corresponding to different training goals [4]:

區間 (Zone)	強度 (Intensity)	最大心率百分比 (% of MHR)	主要訓練效果 (Primary Training Effect)
1	非常輕 (Very Light)	50–60%	幫助熱身與緩和，促進恢復 (Aids in warm-up, cool-down, and recovery)
2	輕度 (Light)	60–70%	提升基礎耐力與脂肪燃燒效率 (Improves basic endurance and fat burning)
3	中等 (Moderate)	70–80%	增強有氧適能與心血管循環 (Enhances aerobic fitness and circulation)
4	困難 (Hard)	80–90%	提高無氧閾值與最大運動表現 (Increases anaerobic threshold and performance)
5	最大 (Maximum)	90–100%	發展最大衝刺能力與速度 (Develops maximum sprint capacity and speed)

心率異常及其健康風險 (Abnormal Heart Rate and Health Risks)

心率異常主要分為心搏過緩（Bradycardia）和心搏過速（Tachycardia）。雖然暫時性的心率變化是正常的，但持續性的異常可能預示著潛在的健康問題。

Abnormal heart rate is mainly categorized into bradycardia (slow heart rate) and tachycardia (fast heart rate). While temporary fluctuations in heart rate are normal, persistent abnormalities may indicate underlying health problems.

心搏過緩 (Bradycardia)

定義： 成人靜息心率持續低於 **60 bpm**。

值得注意： 對於經常鍛煉的運動員來說，靜息心率在40-60 bpm之間可能屬於正常現象。然而，如果心率過慢並伴隨頭暈、極度疲勞、胸痛或昏厥等症狀，則應尋求醫療協助。

潛在風險： 嚴重或未經治療的心搏過緩可能導致心臟無法泵出足夠的富氧血液，引發昏厥、心臟衰竭，甚至心搏停止。

Definition: An adult resting heart rate consistently below **60 bpm**.

Note: For well-trained athletes, a resting heart rate between 40-60 bpm can be normal. However, if a slow heart rate is accompanied by symptoms like dizziness, extreme fatigue, chest pain, or fainting, medical attention should be sought.

Potential Risks: Severe or untreated bradycardia can lead to the heart not pumping enough oxygen-rich blood, causing fainting, heart failure, or even cardiac arrest.

心搏過速 (Tachycardia)

定義： 成人靜息心率持續高於 **100 bpm**。

值得注意： 壓力、焦慮、發燒、脫水或攝入咖啡因和尼古丁都可能導致暫時性心跳加快。但如果靜息心率持續偏高，則需引起重視。

潛在風險： 長期心搏過速會增加心臟負擔，降低泵血效率，可能導致心臟衰竭、中風或因心室顫動引發的猝死。

Definition: An adult resting heart rate consistently above **100 bpm**.

Note: Stress, anxiety, fever, dehydration, or consumption of caffeine and nicotine can cause a temporary increase in heart rate. However, a persistently high resting heart rate requires attention.

Potential Risks: Long-term tachycardia increases the heart's workload and reduces its pumping efficiency, potentially leading to heart failure, stroke, or sudden death due to ventricular fibrillation.

如何改善心率健康 (How to Improve Heart Rate Health)

通過調整生活方式，可以有效改善靜息心率，降低心血管疾病風險。以下是一些關鍵的改善策略：

By adopting a healthier lifestyle, you can effectively improve your resting heart rate and reduce the risk of cardiovascular disease. Here are some key strategies:

1. **規律運動 (Regular Exercise):** 進行規律的有氧運動，如快走、慢跑、游泳或騎自行車，是降低靜息心率最有效的方法之一。建議每週至少進行150分鐘的中等強度運動。
2. **健康飲食 (Healthy Diet):** 採用富含水果、蔬菜、全穀物和瘦肉蛋白的均衡飲食。減少鹽、糖和飽和脂肪的攝入，有助於維持健康體重和血壓。
3. **壓力管理 (Stress Management):** 長期壓力會導致心率升高。嘗試通過冥想、深呼吸、瑜伽或培養個人愛好來減輕壓力。

4. **充足睡眠 (Adequate Sleep):** 確保每晚7-9小時的優質睡眠。睡眠不足會對心血管系統造成壓力，導致心率上升。
5. **戒菸限酒 (Quit Smoking and Limit Alcohol):** 尼古丁和酒精都會直接導致心率加快。戒菸和限制酒精攝入對心臟健康至關重要。
6. **維持健康體重 (Maintain a Healthy Weight):** 肥胖會增加心臟的負擔。通過飲食和運動將體重控制在健康範圍內，有助於降低靜息心率。
7. **Regular Exercise:** Engaging in regular aerobic activities like brisk walking, jogging, swimming, or cycling is one of the most effective ways to lower RHR. Aim for at least 150 minutes of moderate-intensity exercise per week.
8. **Healthy Diet:** Adopt a balanced diet rich in fruits, vegetables, whole grains, and lean proteins. Reducing your intake of salt, sugar, and saturated fats helps maintain a healthy weight and blood pressure.
9. **Stress Management:** Chronic stress can elevate your heart rate. Try to reduce stress through meditation, deep breathing exercises, yoga, or engaging in hobbies.
10. **Adequate Sleep:** Ensure you get 7-9 hours of quality sleep per night. Lack of sleep can put a strain on your cardiovascular system, leading to a higher heart rate.
11. **Quit Smoking and Limit Alcohol:** Nicotine and alcohol both directly increase heart rate. Quitting smoking and limiting alcohol consumption are crucial for heart health.
12. **Maintain a Healthy Weight:** Obesity increases the workload on your heart. Managing your weight through diet and exercise can help lower your RHR.

結論 (Conclusion)

心率是一個動態且個人化的健康指標。了解並監測自己的靜息心率和運動心率，是掌握自身健康狀況、評估體能進步以及預防心血管疾病的重要一環。雖然本報告提供了各年齡層和體能水平的通用參考標準，但更重要的是了解什麼是您個人的「正常」範圍。若發現心率持續異常或伴隨不適症狀，應及時諮詢醫療專業人員，以獲得準確的診斷和適當的建議。

Heart rate is a dynamic and personalized health indicator. Understanding and monitoring your own resting and exercise heart rates are crucial steps in managing your health, evaluating fitness progress, and preventing cardiovascular diseases. While this report provides general reference standards for various ages and fitness levels, it is more important to understand what constitutes a "normal" range for you as an individual. If you notice persistent abnormalities in your heart rate or experience accompanying symptoms, it is essential to consult a healthcare professional promptly for an accurate diagnosis and appropriate advice.

參考資料 (References)

[1] Cleveland Clinic. (2025, September 13). Heart Rate: Normal Rates & What To Know. Retrieved from

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[3] American Heart Association. (2024, August 12). Target Heart Rates Chart. Retrieved from

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